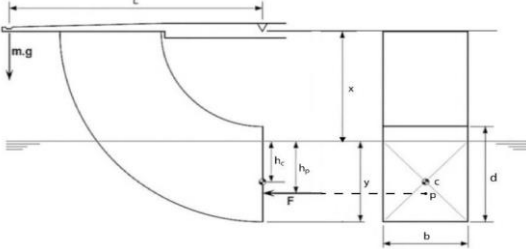
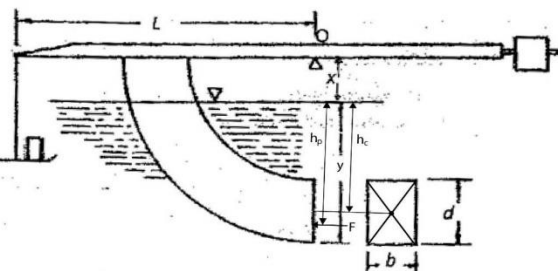


Exp.
01**Determination of the location of the Center of Pressure for a Submerged Plane Surface.**

Formulas:

Partial Immersion ($y < d$):	
	$h_c = y/2$
	$F = \frac{1}{2} \rho g b y^2$
	$h_{pt} = 2/3 y$
	$h_{p(\text{exp})} = \frac{W \cdot L}{F} - x$
Complete Immersion ($y > d$):	
	$h_c = y - \frac{d}{2}$
	$F = \rho g b d (y - \frac{d}{2})$
	$h_{pt} = \frac{d^2}{12(y - \frac{d}{2})} + (y - \frac{d}{2})$
	$h_{p(\text{exp})} = \frac{W \cdot L}{F} - x$

Data Collection

Internal radius of quadrant =

External radius of quadrant =

Width of the rectangular vertical end face, b =

Height of the rectangular vertical end face, d =

No of Obs.	$W = mg$	F	y	x	h_{pt}	$h_{p(\text{exp})}$	h_c
Partial Immersion							
Complete immersion							

Sample Calculation:

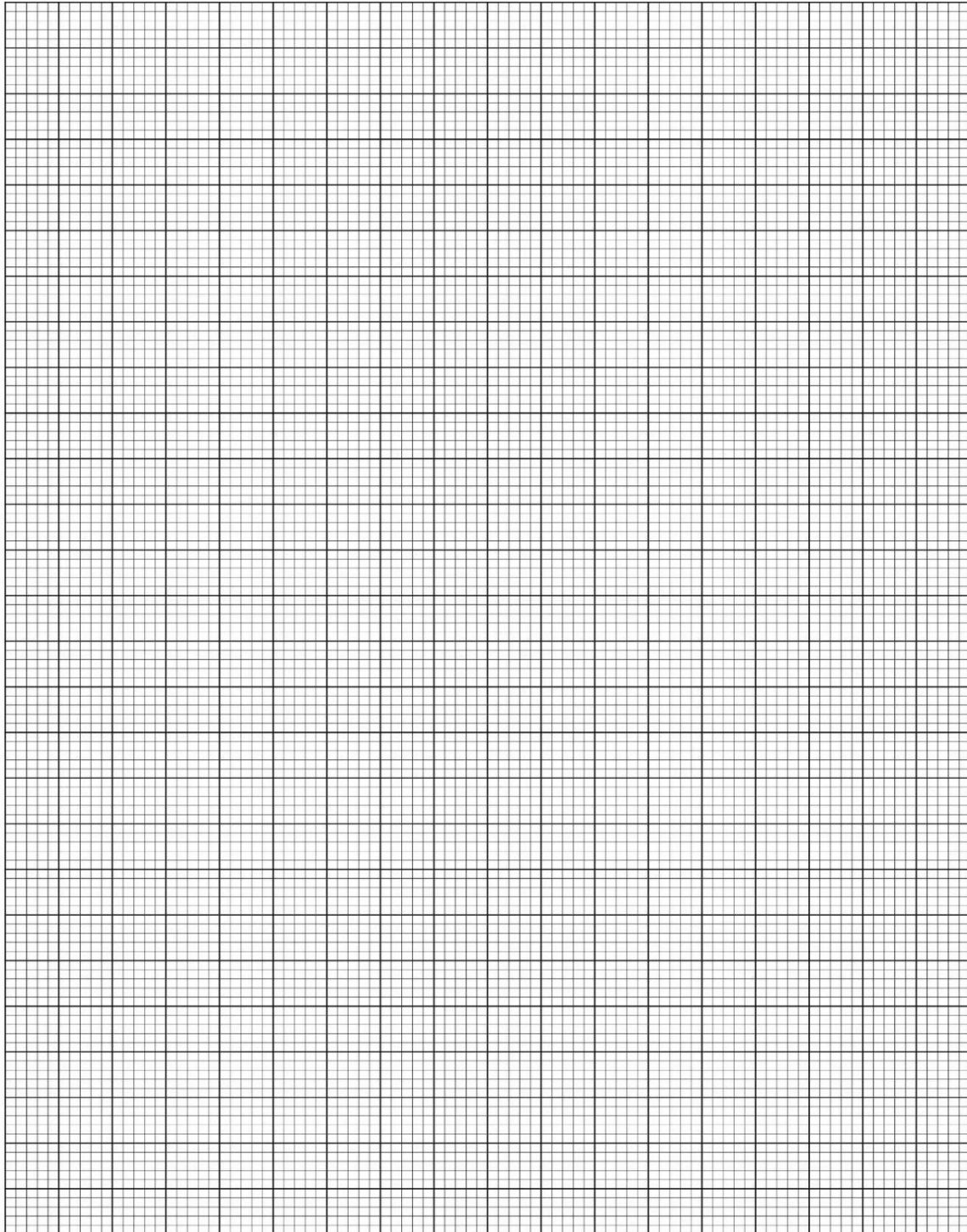
Show a sample calculation for one observation each for partial and complete immersion

Plot:

- Plot $h_{p(\text{exp})}$ vs y and h_{pt} vs y for **Partial Immersion**

Along Horizontal Axis:

Along Vertical Axis:

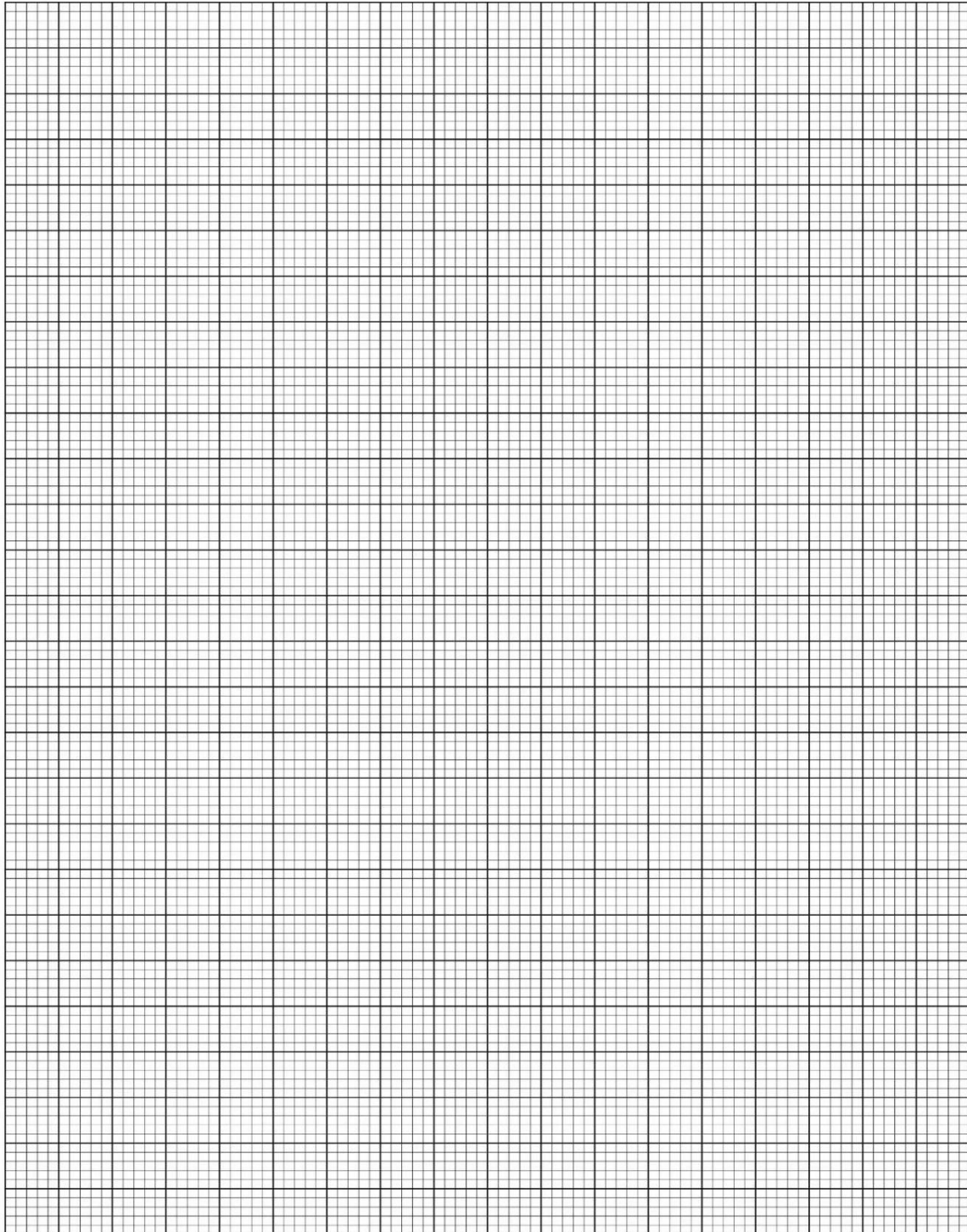


Plot:

- Plot $h_{p(\text{exp})}$ vs y and h_{pt} vs y for **Complete Immersion**

Along Horizontal Axis:

Along Vertical Axis:



Discussion:

- Give reasons for discrepancies between the theoretical and measured values of the results.
- Possible Sources of errors.